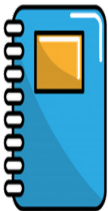




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CSE 3203 – Software Engineering

Lecture 06: Software Processes

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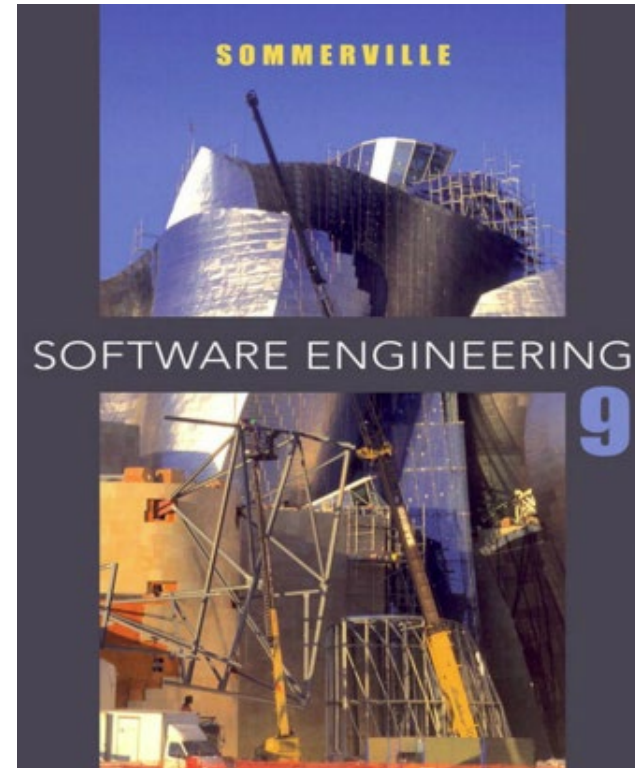
Class Code
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Fall 2025

Book for Reference

- Software Engineering 9th Edition by Ian Sommerville



Mark Distribution



- Mid Semester Exam – 25%
- Final Semester Exam – 40%
- Class Attendance & Performance – 10%
- Course Project – 25%

Is it possible to get A/A+ ?

- Absolutely. Just follow my instructions to the point.

How to fail in this course ?

- Don't follow my instructions.

Instructions



Topics covered

- ✧ Software process models
- ✧ Process activities
- ✧ Coping with change
- ✧ The Rational Unified Process
 - An example of a modern software process.

The software process

- ✧ A structured set of activities required to develop a software system.
- ✧ Many different software processes but all involve:
- ✧ Specification – defining what the system should do;
- ✧ Design and implementation – defining the organization of the system and implementing the system;
- ✧ Validation – checking that it does what the customer wants;
- ✧ Evolution – changing the system in response to changing customer needs.

Software process descriptions

- ✧ When we describe and discuss processes, we usually
 - talk about the activities.
- ✧ Process descriptions may also include:
- ✧ Products: outcomes of a process activity;
- ✧ Roles: responsibilities of the people involved in the process;
- ✧ Pre- and post-conditions: statements that are true before and after a process activity has been enacted or a product produced.

Plan-driven and agile processes

- ✧ Plan-driven: all of the process activities are planned in
 - advance and progress is measured against this plan.
- ✧ Agile: planning is incremental, it is easier to change the process to reflect changing customer requirements.
- ✧ In practice, most practical processes include elements of both plan-driven and agile approaches.

Software process models

✧ The waterfall model

- Plan-driven model. Separate and distinct phases of specification and development.

✧ Incremental development

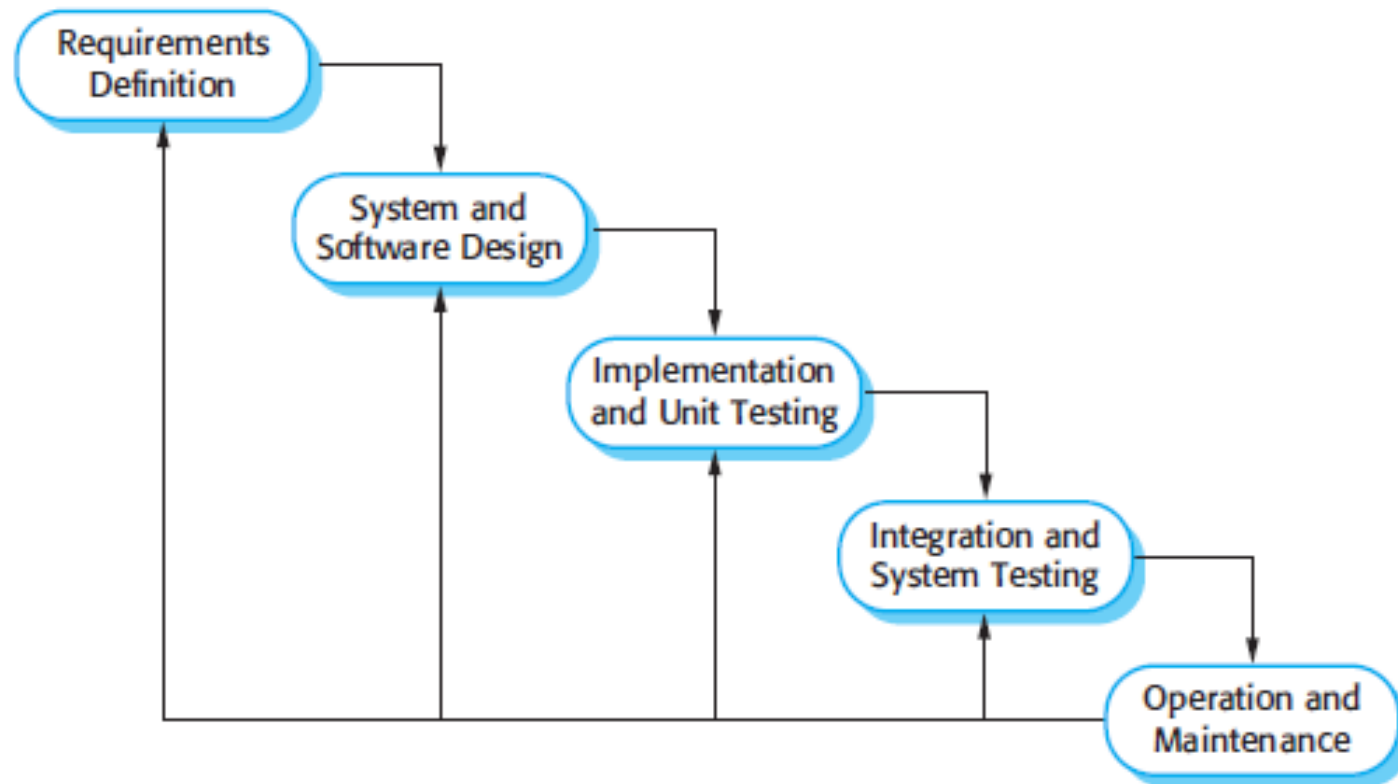
- Specification, development and validation are interleaved. May be plan-driven or agile.

✧ Reuse-oriented software engineering

- The system is assembled from existing components. May be plan-driven or agile.

✧ In practice, most large systems are developed using a process that incorporates elements from all of these models.

The waterfall model



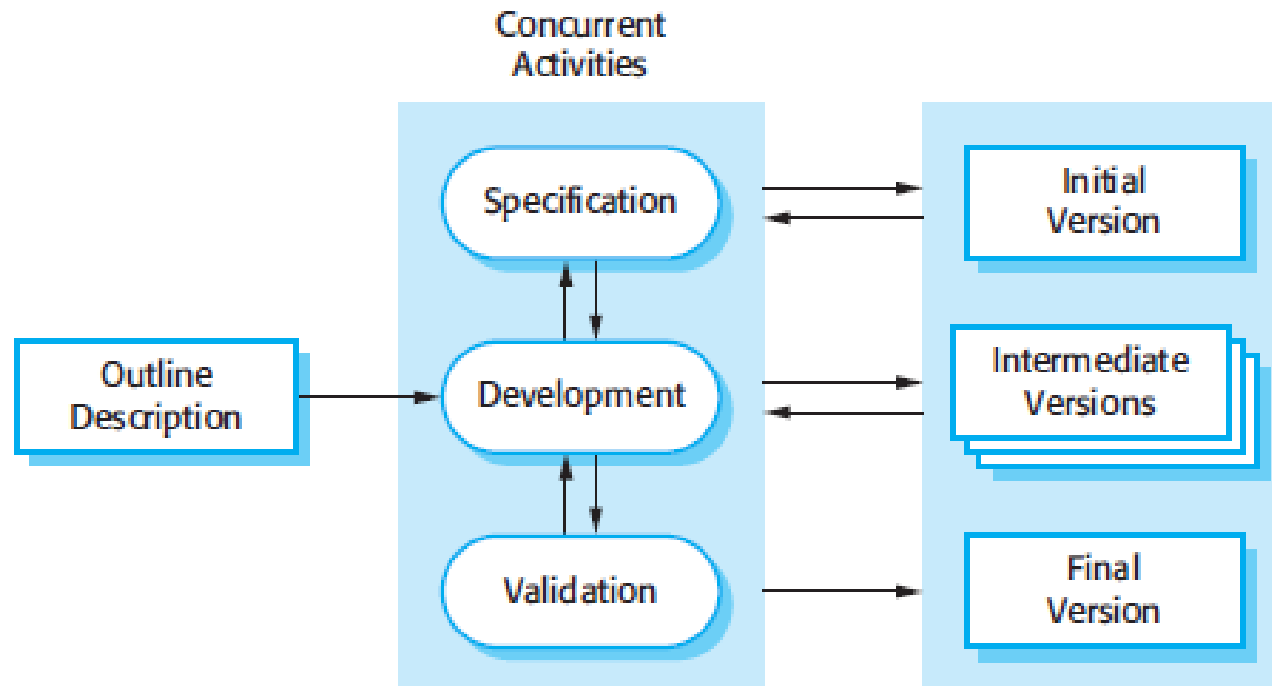
Waterfall model phases

- ✧ There are separate identified phases in the waterfall model:
 - Requirements analysis and definition
 - System and software design
 - Implementation and unit testing
 - Integration and system testing
 - Operation and maintenance
- ✧ The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway. In principle, a phase has to be complete before moving onto the next phase.

Waterfall model problems

- ✧ Inflexible partitioning of the project into distinct stages makes it difficult to respond to changing customer requirements.
 - Therefore, this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process.
 - Few business systems have stable requirements.
- ✧ The waterfall model is mostly used for large systems engineering projects where a system is developed at several sites.
 - In those circumstances, the plan-driven nature of the waterfall model helps coordinate the work.

Incremental development



Incremental development benefits

- ✧ The cost of accommodating changing customer
 - requirements is reduced.
- ✧ It is easier to get customer feedback on the development work that has been done.
- ✧ More rapid delivery and deployment of useful software to the customer is possible.

Incremental development problems

- ✧ The process is not visible.
 - Managers need regular deliverables to measure progress.
- ✧ System structure tends to degrade as new increments are added.
 - Unless time and money is spent on refactoring to improve the software, regular change tends to corrupt its structure.

Reuse-oriented software engineering

- ✧ Based on systematic reuse where systems are integrated from existing components or COTS (Commercial-off-the-shelf) systems.
- ✧ Process stages
 - Component analysis;
 - Requirements modification;
 - System design with reuse;
 - Development and integration.
- ✧ Reuse is now the standard approach for building many types of business system

Reuse-oriented software engineering

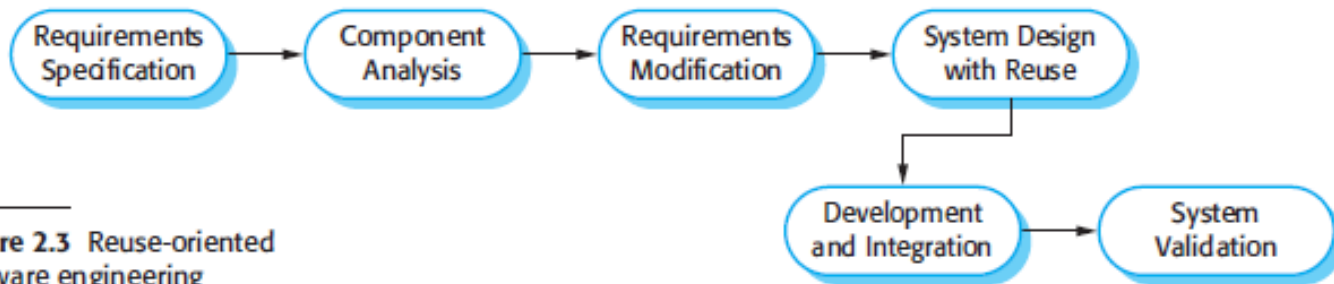


Figure 2.3 Reuse-oriented software engineering

Types of software component

✧ Web services that are developed according to service

- standards and which are available for remote invocation.

✧ Collections of objects that are developed as a package to be integrated with a component framework such as

- .NET or J2EE.



Thank You

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